Sherwin Williams Recommendation System

A Data-Driven Approach to Paint Selection

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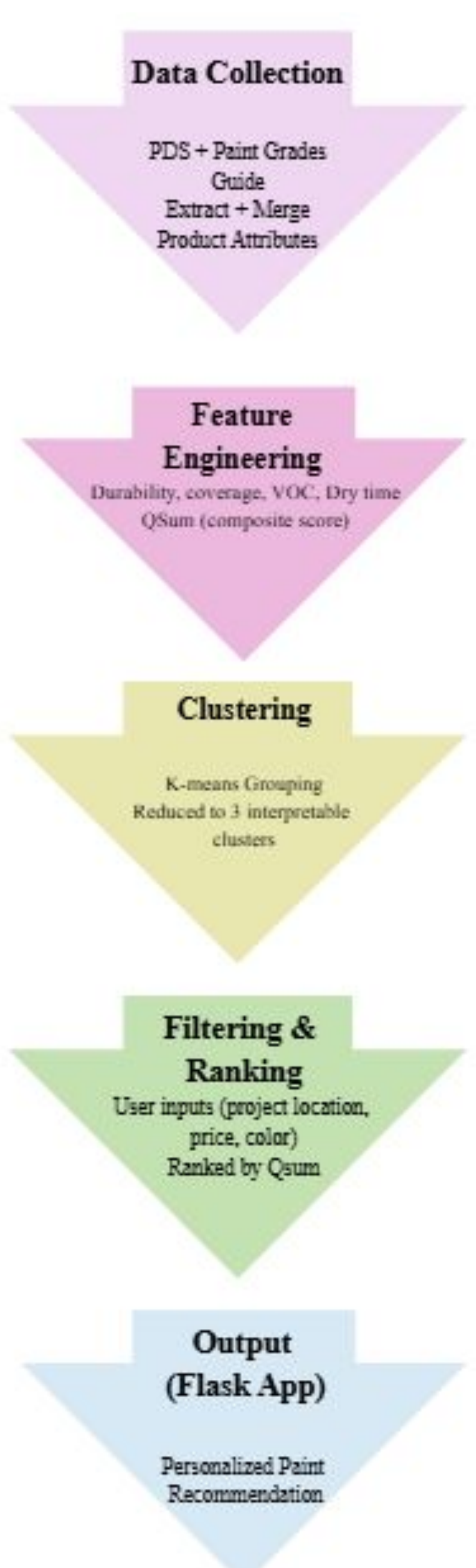
Background & Overview

- Builds a paint recommendation system for Sherwin-Williams products using structured data, feature engineering, and clustering.
- Combines multiple datasets with user-driven filters to deliver tailored recommendations through a Flask web app.
- After the 1978 ban on lead-based paint by the Consumer Product Safety Commission, product options expanded significantly (mold & mildew resistance, low-VOC, antimicrobial coatings, etc).
- This growth in industry created a complex decision-making environment, which this system simplifies into clear, user-friendly recommendations.

Methodology

System Pipeline



Public Sherwin-Williams PDS and Paint Grades Guide were used to gather attributes such as durability, coverage, VOC levels, drying time, and mold resistance. Data was cleaned and merged into a unified dataset.

Key attributes were structured into usable features, including performance metrics and technical details. A composite score (QSum) was created to represent overall product performance.

K-means clustering grouped similar paint products based on engineered features. Initial clusters were refined into three interpretable categories for clarity.

User-defined filters (project location, price range, and color type) narrowed the dataset. Results were ranked using QSum to return the most relevant products.

The system was developed as a Flask web application, allowing users to input preferences and receive tailored paint recommendations.

Results

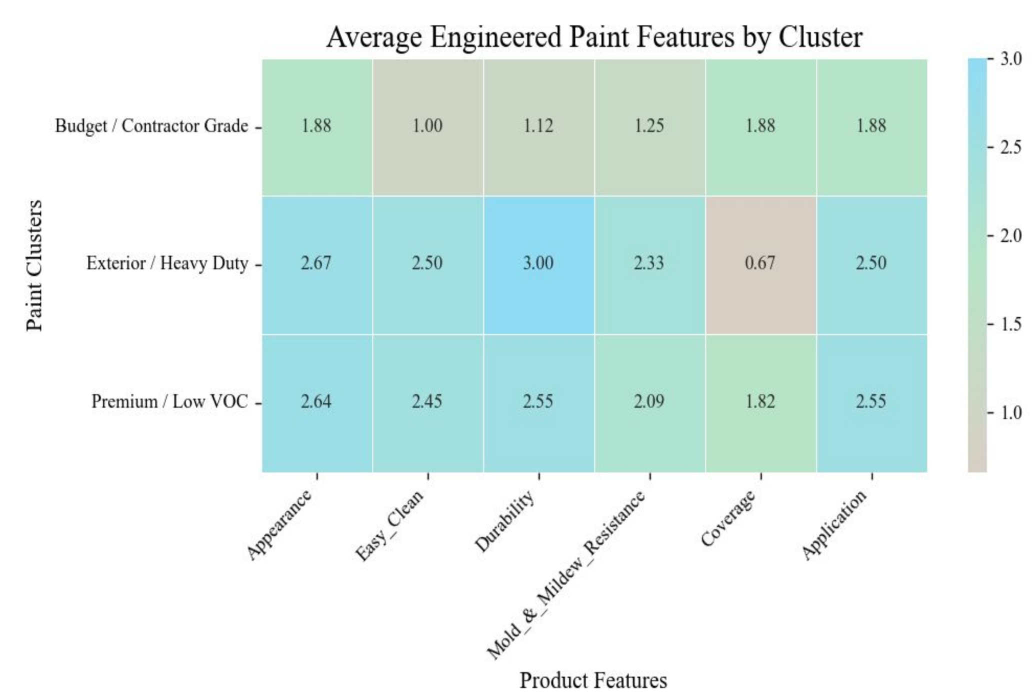


Figure 1: Heatmap of average engineered features across K-means clusters. Reveals distinct product groupings based on performance characteristics.

- **Exterior / Heavy Duty**
  Products with stronger exterior use characteristics
- **Budget / Contractor Grade**
  Lower overall performance scores and more basic options
- **Premium / Low VOC**
  Higher-performing products with lower VOC levels
- **Premium / High Durability**
  Strong overall performance with emphasis on durability

Figure 2: Product count before and after filtering. Demonstrates how user inputs reduce the search space and improve recommendation focus.

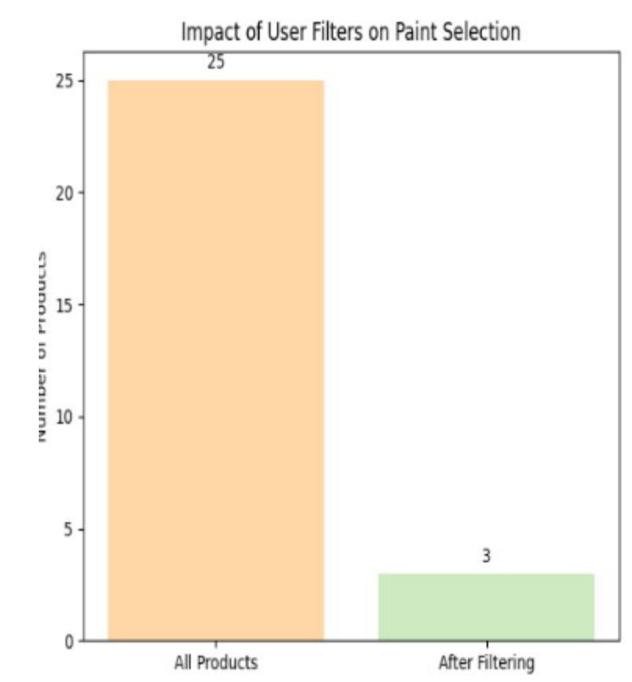
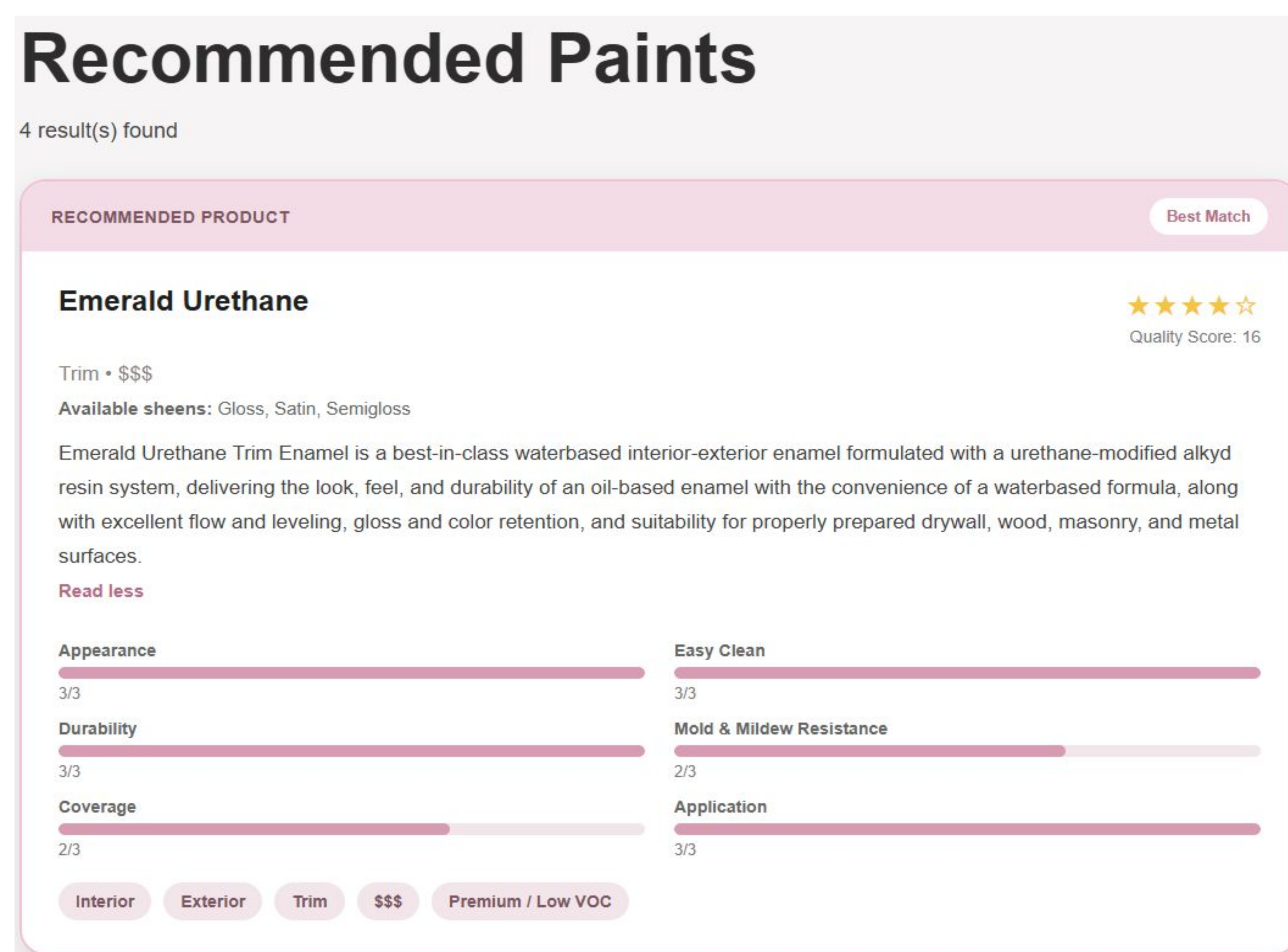


Figure 3: Example output from the Flask application showing recommended products ranked by performance (QSum) based on user-defined filters.



The results highlight the challenge of distinguishing signal from noise in product data. Although paint products often appear similar, the use of structured features and clustering reveals meaningful differences in performance and intended use. When combined with user-driven filtering, the system provides a clearer and more efficient way to navigate a complex product space.

Discussion & Conclusion

The system generates personalized paint recommendations based on user input, returning products that match selected criteria such as use case, price range, and color type. Results are ranked using a performance based score (QSum), ensuring higher-quality products appear first. Clustering further enhances the output by grouping similar paints, helping users compare options within meaningful categories and better understand trade-offs between products.

Implications:

- Transforms complex manufacturer data into user-friendly decision support.
- Reduces reliance on heavy product descriptions and technical documentation.
- Demonstrates how combining structured data, filtering, & modeling improves recommendation quality.

Limitations

- Clustering often reflects existing product similarities, offering limited additional differentiation.
- Recommendations depend on available data and may not capture all real-world factors (user preference, brand familiarity).
- The dataset is static and does not automatically update with newly released Sherwin-Williams products (recently introduced products like “Forte” and “Symmetry”).

Future Directions

- Incorporate user preference weighting (prioritize durability vs cost).
- Expand dataset to include more products and real user feedback.
- Explore alternative models beyond K-means for more advanced product classification

References & Github